AI INFRASTRUCTURE / TECHNICAL REPORT

Memory-Bounded Teacher Scoring

A contract-preserving readout for a verl FSDP2 Worker

81.05% lower peak allocated

At 8,192 projected rows, the complete Worker call used 3,596.6 MiB with dense readout and 681.6 MiB with row chunking. All 262,144 candidate logprobs and 8,192 ordered Teacher Top-K rows matched the dense reference exactly.

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Evaluated system	Qwen2.5-0.5B-Instruct, BF16, one GPU, FSDP2 CPU offload
Readout	$K = 32$, 256 rows per chunk, allocator fraction 0.50
Evaluation baseline	`ad395e94923bcf65fff4bc1eeb2e0e1505fd4f91`

The percentage is the PyTorch allocator's peak allocated during a complete Worker call, not a whole-device VRAM reduction.

Executive summary

A teacher can return a compact set of scores while still materializing logits for every projected row and every vocabulary entry. This project moves row chunking ahead of the Qwen2 LM head inside the normal FSDP2 forward. Each tile produces full-vocabulary-normalized scores at Student candidate IDs and the Teacher's ordered Top-K; only compact outputs are retained. The adapter is opt-in and preserves the tied head, module hooks and output layout of the pinned verl Worker.[3]

On an RTX 5070 Laptop GPU, the $N = 8192$ experiment reduced PyTorch peak allocated by 81.05% under the stated allocator fraction. A separate direct Worker-to-Student fixture matched the pinned reverse-KL loss, checked gradients and one restricted final-RMSNorm update. GitHub's CPU regression workflow passed 171 tests, with two full Trainer-runtime parity tests deselected. The result is a single-GPU experimental execution-path optimization; the time measurements are descriptive and do not establish a general speedup.

1 The engineering problem

Let N be the number of projected model rows, V the vocabulary size and K the number of requested candidates per row. A dense head first creates $N \times V$ logits, although the response contains only $O(NK)$ values and IDs. Candidate-only normalization would change the contract: the consumer needs candidate logprobs normalized over the full vocabulary as well as the Teacher's own Top-K.

Row chunking is an established technique; the contribution here is its integration with a real Worker and its scoring and consumer interfaces. Related selected-logit implementations and per-row scoring interfaces are prior work.[1, 2]

2 System overview

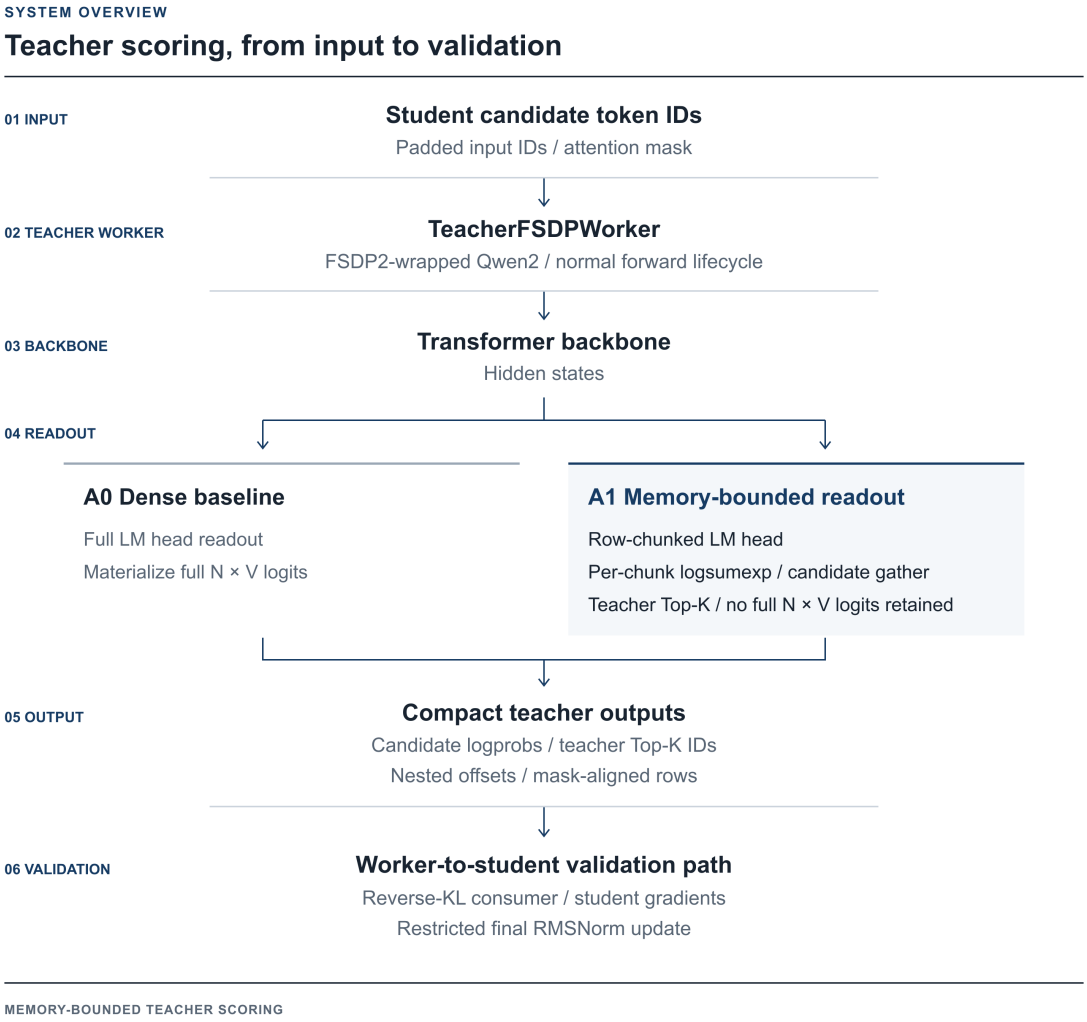


Figure 1. Dense and row-chunked readout paths share the Teacher execution and compact-output contract. The last stage depicts a separate validation fixture.

3 Method and implementation

3.1 The scoring contract

For hidden row h_t, LM-head weights W, bias b and candidate i, the returned score is

$$z_{t,i} = h_t^{\mathsf{T}} W_i + b_i, \qquad \log p_{t,i} = z_{t,i} - \log \sum_{j \in \mathcal{V}} e^{z_{t,j}}. \tag{1}$$

The normalizer ranges over the full vocabulary. Candidate IDs are ordered and may repeat; the returned values retain that order. The Worker also returns Teacher Top-K IDs over the full vocabulary, with nested offsets and valid-row order preserved.

3.2 Bounded readout inside FSDP2

The adapter is installed before FSDP wrapping, retaining parameter objects and the tied input/output weight. A request enters through the normal outer FSDP module call. The backbone runs once; its hidden rows are then passed through the LM head as a module in tiles of at most B rows. For each tile, logits are promoted to FP32 for log-sum-exp and candidate gather, and Teacher Top-K IDs are retained. Tiles are discarded without assembling a full $N \times V$ result.

The live logits tile scales with BV instead of NV; compact outputs still scale with NK, and backbone, model and FSDP allocations remain. This changes allocation behavior rather than eliminating the full-vocabulary projection work. When an unbudgeted tile covers all rows, the implementation uses the native dense forward; an explicit readout budget is an estimate for that portion of the computation, not a whole-device limit.

3.3 Padding and consumer alignment

Candidate IDs are scattered into padded model rows with the prepared attention mask, then compact outputs are restored to valid-row order. The adapter does not add a causal shift to candidate IDs. The consumer fixture applies the response mask at the appropriate prediction positions and passes the actual Worker outputs to the pinned loss.

4 Evaluation

4.1 Setup

The final matrix used Qwen2.5-0.5B-Instruct in BF16 on one RTX 5070 Laptop GPU, PyTorch 2.14.1+cu130, FSDP2 with CPU offload, SDPA, $K = 32$, $B = 256$ and allocator fraction 0.50. Inputs and candidate IDs were synthetic. For each length, four fresh processes ran Dense–Chunked–Chunked–Dense; each process performed one warmup and five measured calls. N denotes projected rows.

N	Mode	Alloc. MiB	Reserved MiB	p50 ms	p95 ms
1024	Dense	1517.1	2644	210.4	288.0
1024	Chunked	665.3	1124	159.4	187.0
4096	Dense	2409.4	2944	322.3	414.4
4096	Chunked	672.3	1130	318.8	327.5
8192	Dense	3596.6	3846	570.8	599.6
8192	Chunked	681.6	1126	626.3	944.6

Table 1. Per-call PyTorch allocator peaks and synchronized warm Worker spans. Each p50/p95 is computed over ten measured calls.

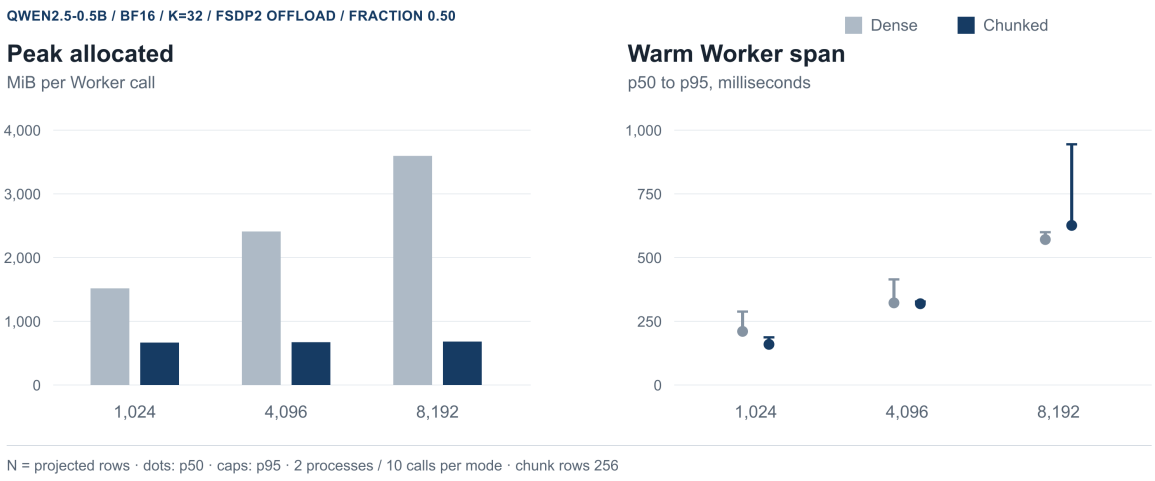


Figure 2. Peak allocated and the p50-to-p95 span; reserved memory is shown in Table 1.

5 Correctness and interpretation

At $N = 8192$, all 262,144 candidate logprobs and all 8,192 ordered Teacher Top-K rows matched the Dense reference exactly. Across the final matrix and chunk-size sweep, 96 saved compact outputs were reloaded and compared against same-length Dense references. The comparison includes output identity, shape and offsets.

The $N = 8192$ peak-allocated decrease from 3,596.6 to 681.6 MiB is 81.05%. It is a per-call PyTorch allocator measurement, not an estimate of total device VRAM. Under the same fraction, Dense succeeded at 8,704 rows and ran out of allocator memory at 9,216; Chunked ran at 32,768, the largest tested point. A same-length Dense correctness reference was not available for 32,768.

The synchronized timings include diagnostic hooks and return of nested outputs to CPU. At 8,192 rows, Dense p50/p95 was 570.8/599.6 ms and Chunked was 626.3/944.6 ms. Process variation and host-clock differences limit interpretation; these measurements do not establish a general latency improvement.

5.1 Worker-to-Student validation

In a separate two-sequence fixture, actual Dense and Chunked Worker outputs fed the pinned reverse-KL consumer. The loss, Student logits gradients, final RMSNorm gradients and one restricted AdamW update matched. The fixture used $K = 8$, 25 active response-prediction rows and updates only the Student's final RMSNorm weight. An earlier scalar-loss match concealed gradient differences; aligning the conversion graph with the pinned consumer removed those differences in the recorded fixture. This is a direct integration check, not a full RL training run.

6 Current scope and future work

The evaluated path covers tied-head Qwen2 on one GPU with FSDP2. The next stage is to extend model coverage, support packed variable-length inputs, validate FSDP1 and multi-GPU/tensor-parallel execution, and integrate an end-to-end RL trainer. The initial selected-logit-only fusion proposal remains NO-GO for the complete scoring contract; historical negative findings remain in the project results.

7 Reproducibility and evidence

The evaluated code baseline is `ad395e94923bcf65fff4bc1eeb2e0e1505fd4f91`; the Worker integration targets verl `87cd055e872f9fdd7b976fcd797a0934b6dba265`, and the model revision is `7ae557604adf67be50417f59c2c2f167def9a775`. The dedicated Linux/Python 3.12 Worker lock fixes 105 dependencies. The repository's reproduction guide specifies the setup, model hashes, commands and experiment boundaries.

The public results summary contains the matrix values, and the evidence manifest binds archived measurements to SHA-256 hashes. The GitHub CPU job ran at the baseline revision and reported 171 passed, 2 deselected and no skipped tests; it checks CPU regressions and patch applicability rather than running the GPU benchmark.

- Experimental reproduction guide
- Worker benchmark summary and evidence manifest
- GitHub Actions CPU run

8 Conclusion

Moving the row boundary ahead of the LM head gives this Teacher Worker a smaller live readout allocation while preserving the outputs required by its consumer. The measured single-GPU result, exact output comparisons and restricted Student update provide a concrete basis for the next integration stage.

References

[1] Jordan Pettyjohn. IndexedLogits: indexed selected-logit CUDA implementation, 2026. URL `https://github.com/pettyjohnjn/indexed_logits/tree/6020cb446cde8cb2b74a8fdf90db905a320936c3`. Source inspected 2026-10-28; no project runtime reproduction.

[2] SGLang contributors. PR 38119: Add sparse per-position requested-token scoring, 2026. URL `https://github.com/sgl-project/sglang/pull/38119`. Source inspected 2026-10-28; no project runtime reproduction.

[3] verl contributors. Teacher scoring and student-top-K distillation, PR 6920, 2026. URL `https://github.com/verl-project/verl/pull/6920`. Experiments use source revision 87cd055e872f9fdd7b976fcd797a0934b6dba265 and a local project patch; no upstream acceptance is implied.